

WARRENSBURG SKYHAVEN, MO, USA

WMO: 722367

Lat: 38.783N

Lon: 93.800W

Elev: 798

StdP: 14.28

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 00363

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.9	(c) 10.2	(d) -4.0	(e) 4.6	(f) 7.0	(g) 0.9	(h) 5.9	(i) 13.2	(j) 23.5	(k) 40.1	(l) 21.0	(m) 34.1	(n) 5.7	(o) 290	(p) 0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	94.6	73.3	90.7	74.6	88.5	74.3	78.8	87.2	77.4	86.6	75.9	84.8	8.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.8	144.1	84.4	73.5	128.2	82.2	72.8	125.1	81.4	43.1	86.2	41.2	86.1	39.8	84.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	18.5	16.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-3.4	81.0	7.8	2.0	-9.0	82.4	-13.6	83.6	-18.0	84.7	-23.7	86.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.0	30.2	34.4	46.1	54.4	64.3	74.1	76.5	74.7	68.3	56.2	44.6	34.3
	DBStd	18.51	11.39	11.45	11.13	9.37	8.38	5.91	5.66	5.53	7.45	9.32	9.86	9.90
	HDD50	2096	619	453	208	59	5	0	0	0	44	213	495	
	HDD65	4910	1079	858	592	335	118	7	1	3	49	301	614	953
	CDD50	3904	5	16	88	192	449	723	823	765	549	237	50	7
	CDD65	1241	0	0	7	17	98	280	359	303	147	29	1	0
	CDH74	11603	0	1	48	150	781	2735	3626	2760	1270	228	4	0
	CDH80	4336	0	0	2	19	186	1026	1515	1100	444	44	0	0
(14) Wind	WSAvg	7.0	7.9	8.4	8.8	8.8	7.0	6.5	4.9	4.7	5.1	6.8	7.8	7.8
(15) Precipitation	PrecAvg	41.9	1.3	1.9	2.8	4.6	5.6	6.1	4.6	3.3	4.3	3.4	2.5	1.8
	PrecMax	56.0	3.7	4.9	6.3	12.2	12.3	11.3	14.0	8.9	11.1	8.1	8.2	4.5
	PrecMin	29.1	0.3	0.2	0.9	1.1	0.7	2.4	0.7	1.2	0.6	0.5	0.3	0.2
	PrecStd	8.0	0.8	1.2	1.4	2.1	3.1	2.4	3.0	1.8	3.2	1.8	2.0	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	72.0	79.1	82.1	88.3	96.6	100.3	97.5	92.8	85.5	72.9	65.8
		MCWB	54.8	58.0	62.0	65.2	72.1	73.9	71.6	75.6	72.5	70.1	60.1	59.4
	2%	DB	59.0	64.4	73.3	78.9	85.5	90.9	95.0	92.8	89.7	80.8	69.8	59.4
		MCWB	49.2	53.0	60.1	62.8	69.9	75.1	72.3	73.8	73.0	65.1	58.9	52.1
	5%	DB	52.3	59.5	70.0	73.4	81.5	89.7	90.7	90.0	86.0	75.5	64.0	54.1
		MCWB	44.5	50.6	58.1	60.4	68.5	74.7	75.3	74.3	72.4	62.6	55.2	48.1
	10%	DB	45.7	52.4	64.2	71.5	79.0	86.3	88.2	86.4	81.4	72.1	60.8	50.1
		MCWB	40.0	46.0	54.7	59.4	67.1	73.6	75.2	73.2	69.3	61.1	51.9	44.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.0	60.1	65.8	68.1	74.7	79.8	81.0	80.4	76.3	71.3	64.0	60.4
		MCDB	64.2	68.0	76.3	79.9	84.3	88.6	88.8	90.3	86.8	81.9	71.2	64.6
	2%	WB	49.9	56.7	63.1	65.7	72.6	78.0	79.3	78.1	74.4	68.5	60.2	54.3
		MCDB	57.1	63.3	72.4	75.0	81.7	86.7	87.7	87.7	85.4	75.5	65.8	58.3
	5%	WB	45.2	51.3	59.8	63.4	70.7	76.2	77.7	76.5	73.0	66.2	57.0	49.0
		MCDB	51.5	59.7	67.6	70.9	78.7	85.8	86.2	85.8	83.4	72.7	63.6	52.9
	10%	WB	40.3	45.8	55.9	60.4	68.6	74.5	76.1	74.6	71.6	63.2	53.2	44.1
		MCDB	46.3	52.4	63.3	68.6	76.3	83.8	84.3	83.3	80.1	69.1	59.2	49.1
(35) Mean Daily Temperature Range	5% DB	MDBR	17.5	18.0	18.8	20.1	18.8	19.2	18.8	19.6	20.6	21.5	19.4	16.3
		MCDBR	25.7	28.3	25.3	24.7	21.8	21.4	21.4	23.7	22.0	25.0	25.0	24.0
		MCWBR	17.3	19.1	13.7	13.6	10.6	9.1	7.6	8.7	9.1	12.3	16.2	17.5
	5% WB	MCDBR	24.3	27.2	22.0	20.3	19.3	19.1	17.6	19.8	20.0	19.9	21.5	22.3
		MCWBR	17.5	19.9	13.5	13.2	10.9	9.8	9.0	9.2	9.4	11.6	16.3	19.0
(40) Clear-Sky Solar Irradiance	taub		0.297	0.310	0.337	0.389	0.428	0.429	0.444	0.446	0.389	0.347	0.307	0.292
	taud		2.461	2.427	2.408	2.279	2.233	2.262	2.236	2.239	2.356	2.455	2.512	2.497
	Ebn at Noon		280	291	292	282	272	270	265	262	272	273	275	274
	Edh at Noon		25	30	34	41	44	43	44	43	36	29	24	23
(44) All-Sky Solar Radiation	RadAvg		692	928	1215	1559	1802	2040	2037	1794	1524	1084	752	597
	RadStd		77	106	108	101	160	164	126	99	98	116	78	57

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page